JACS
JOURNAL OF THE AMERICAN CHEMICAL SOCIETY
pubs.acs.org/JACS
Communication

Construction of Interlayer Conjugated Links in 2D Covalent Organic Frameworks via Topological Polymerization

Yuhao Zhu, Pengpeng Shao, Linyu Hu, Chao Sun, Jie Li, Xiao Feng,* and Bo Wang

Cite This: J. Am. Chem. Soc. 2021, 143, 7897−7902

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Two-dimensional covalent organic frameworks (2D COFs) are well-defined polymeric sheets that usually stack in an eclipsed mode via van der Waals forces. Extensive efforts have been made to manipulate interlayer interactions, yet there still lack a way to construct conjugated connections between adjacent layers, which is important for (opto)electronic-related applications. Herein, we report an interlayer topological polymerization strategy to transform the well-organized diacetylene columnar arrays in three different 2D COFs (TAPFY-COF, TAPB-COF, and TAPP-COF) into conjugated enyne chains upon heating in the solid state. The resultant COFs (COF-P) with retained high crystallinity possess broadened absorption bands and narrowed band gaps. The newly formed conjugated chains provide extra charge carrier pathways through direct π-electron delocalization. As a proof-of-concept, after topological polymerization, the conductivity of the TAPFY-COF film achieves 2.8×10^{-4} S/cm without doping, and the photothermal, photoacoustic, and oxygen reduction catalytic performance of TAPP-COF is significantly improved.

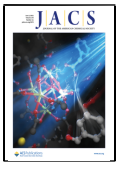
Covalent organic frameworks (COFs) are organized by organic building blocks via covalent bonds, featuring a two- or three-dimensional polymeric backbone with a highly crystalline and porous nature.[1−9] Particularly, 2D COFs are usually constructed by aromatic monomers combined with dynamic covalent bonds to confine the chain growth within 2D planes, forming sheet-like structures with long-range order. These sheets aggregate in the *z*-axis due to interlayer van der Waals interactions, which allows controlling their secondary and tertiary structures for the construction of macromolecule assemblies.[10−18] The stacked structures lead to the formation of efficient open channels for selective substance storage and transport, and the functional moieties can be easily incorporated onto the backbones or into the channels. Moreover, conjugated planar layers provide a potential charge-transport path, and the rich chemical toolbox enables the fine manipulation of the molecular orbitals distributed in the COFs. These features make 2D COFs promising for (opto)electronic devices and photo-/electrocatalysis applications.[19−28] Although the planar sheets in the 2D COFs are supposed to adopt a high conjugation degree to facilitate charge transport, the UV−vis absorption wavelength and the bandgap of the established COFs indicate their conjugation is indeed limited.[29−32] This is due to the rotation of the aromatic rings caused by the steric hindrance between the neighboring moieties. It makes the charge carrier conduction in COFs mainly depend on the through-space pathway offered by $\pi-\pi$ interactions.[33,34] Construction of a through-bond conjugated pathway between the layers may provide an opportunity to further boost their photo and electrical properties and functions.[35−37]

Solid-state topological polymerization (SSTP) provides a great chance to covalently link the adjacent layers in 2D COFs that usually adopt an eclipsed stacking mode with short interlayer distances less than 4 Å.[38−44] Pioneering works have been reported recently to connect the COF layers with cyclobutane links or concavo-convex-shaped dimers through [2+2] and [4+4] cycloaddition, respectively.[45,46] The ion conduction and fluorescence properties of the COFs have been adjusted in these works. Unfortunately, the newly formed connections are not conjugated and even bring a decrement of aromaticity.

Herein, we intended to employ the radical SSTP of diacetylene to achieve the conjugate between adjacent layers in the 2D COFs (Figure 1a). We first used density functional theory (DFT) to theoretically predict the change of band structure and electron cloud distribution in the frontier orbitals of an imine-linked COF (TAPFY-COF) constituted by 4,4′-(buta-1,3-diyne-1,4-diyl)dibenzene and 1,3,6,8-tetrakis-(phenyl)pyrene. In consideration of the atom distance, the SSTP polymerized structure was constructed in a C1-to-C1 and C4-to-C4 connection mode (Figure S4).[47] Indeed, after interlayer polymerization, the resultant framework (denoted as TAPFY-COF-P) shows a significantly narrowed bandgap, and a new conduction pathway through the enyne chain has been formed. Therefore, we experimentally prepared TAPFY-COF and further conducted SSTP. The polymerization process can be easily triggered by heating and can achieve quantitative yield with atom economy.

Crystal structure optimization by force-field calculations suggests that TAPFY-COF adopts ordered 1D rhombic channels and an AA stacking structure with a slight layer

Received: March 18, 2021
Published: May 19, 2021



Downloaded from pubs.acs.org/jacsat/article-pdf/143/21/7897/3884079/ja1c02932.pdf by UNIV DE VALENCIA user on 15 October 2026

© 2021 American Chemical Society

https://doi.org/10.1021/jacs.1c02932
J. Am. Chem. Soc. 2021, 143, 7897−7902

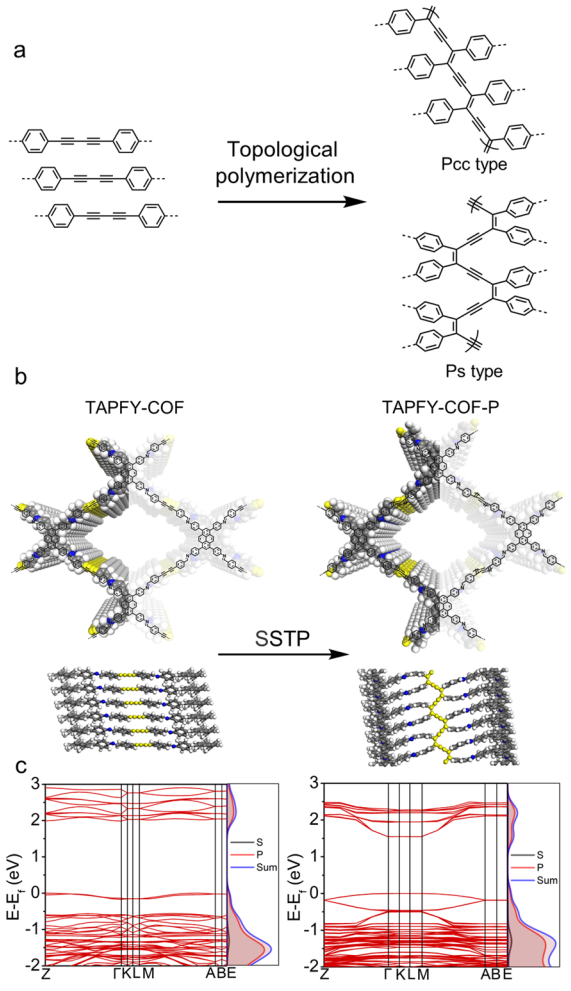


Figure 1. (a) Schematic representation of the two possible polymerization routes of diacetylene. (b) Structures of TAPFY-COF before and after SSTP. (c) Band structure of TAPFY-COF and TAPFY-COF-P.

offset (Figure 1b, Table S2). In addition, the phenyl ring adjacent to the pyrene unit is obviously rotated with a dihedral angle of 41.6° due to the steric hindrance of the neighboring hydrogen atoms. The intermolecular distance between the diacetylene groups is 3.85 Å. After polymerization, the layers are linked by polymeric chains with alternating double−triple bonds (Figure 1b); correspondingly, the force-field optimized crystal structure of TAPFY-COF-P shows a shrinkage of the crystal unit cell (Table S3).

The band structure (Figure 1c) and frontier orbitals (Figures S5 and S6) of TAPFY-COF and TAPFY-COF-P were calculated by DFT. TAPFY-COF exhibits a relatively flat highest occupied molecular orbital (HOMO) band and lowest unoccupied molecular orbital (LUMO) band, indicating that the effective mass in TAPFY-COF is larger and its in-plane and out-of-plane charge transport is limited. This is due to the broken conjugation in the distorted planar layers. In comparison, TAPFY-COF-P displays obvious band dispersion in the out-of-plane direction, revealing more effective charge transport through the enyne-linked layers.[48] After SSTP, the bandgap narrows from ∼2.0 to ∼1.5 eV due to the formation of conjugated enyne chains, which is beneficial for the increase of intrinsic carrier concentration (Figure 1c).[49] In addition, the HOMO wave functions of the two COFs are both localized mainly on the pyrene knots (Figure S5). In contrast to being distributed throughout the framework of TAPFY-COF, the LUMO wave function is localized on the interlayer conjugated chains in TAPFY-COF-P (Figure S6). This indicates that more significant charge separation in the excited state is likely to occur after polymerization. The above results demonstrate that the formation of conjugated enyne chains facilitates the charge transfer between the layers as well as the charge separation in the layers.

Next we synthesized TAPFY-COF under solvothermal condition in an isolated yield of 90%. Thermogravimetric analysis (TGA) reveals that TAPFY-COF showed no weight loss up to 500 °C under N_2 atmosphere (Figure 2a). Based on the differential scanning calorimetry (DSC) and variable-temperature diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) results (Figures S7 and S10), the subsequent topology polymerization was conducted under 350 °C for 6 h in the solid state to achieve complete reaction and avoid solvent-induced changes of interlayer packing.[50−52] Further TGA performed under the same polymerization conditions showed that the carbonization process did not happen during the SSTP procedure (Figure S8).

The successful syntheses of TAPFY-COF and TAPFY-COF-P are proved by Fourier transform infrared (FT-IR) spectroscopy, ^{13}C cross-polarization/magic angle spinning solid-state nuclear magnetic resonance (^{13}C CP/MAS ssNMR) spectroscopy, elemental analyses (EA), powder X-ray diffraction (PXRD), and N_2 adsorption measurements. Comparing with TAPFY-COF, a new peak appears around 1510 cm^{-1}, assigned to the characteristic vibration of C═C in the enyne unit, and the C═N vibration is retained in the FT-IR spectra of TAPFY-COF-P. Moreover, after SSTP, the C═C stretch of the aromatic ring groups around 1580 cm^{-1} shifts to a higher wavenumber, around 1605 cm^{-1} (Figure 2b, Figure S9), indicating the obvious change in the chemical environment of the aromatic rings resulting from a high degree of transformation of the diacetylene groups.

From the ^{13}C CP/MAS ssNMR spectra (Figure 2c), the completion of diacetylene polymerization is evidenced by the disappearance of the two signals at around 75 and 80 ppm assigned to the carbons with different chemical environments in the diacetylene group (Figure 2b,c), revealing the fully conversion of diacetylene. The ssNMR spectrum of TAPFY-COF-P becomes less resolved, which reveals that the carbon atoms have similar but not exactly chemical equivalent environments caused by the imperfectness in the enyne chains. The EA analyze results shows that the elemental contents (C, H, and N) do not change much after SSTP and are close to the theoretical calculated values (Table S1). Compared with the UV−vis diffuse reflectance spectra of TAPFY-COF, the absorption band of TAPFY-COF-P is significantly broadened and a typical $\pi-\pi^*$ transitions absorption peak of polydiacetylene appears around 570 nm (Figure 2g).[53−55] In addition, the experimental deduced band gaps are 2.25 and 1.54 eV for TAPFY-COF and TAPFY-COF-P, respectively, which are agree with the theoretical calculations.

PXRD and theoretical simulations were carried out to deduce their crystal structure. TAPFY-COF shows prominent peaks with 2θ at 2.74, 5.62, 8.46, and ∼23.96°, which can be assigned to (110), (220), (420), and (001) facets, respectively (Figure 2d). The Pawley refinement result, as well as the simulated PXRD pattern obtained by using the optimized unit cell, matches the experimental results (Figure S11). TAPFY-

Downloaded from pubs.acs.org/jacsat/article-pdf/143/21/7897/3884079/ja1c02932.pdf by UNIV DE VALENCIA user on 15 October 2026

https://doi.org/10.1021/jacs.1c02932
J. Am. Chem. Soc. 2021, 143, 7897−7902

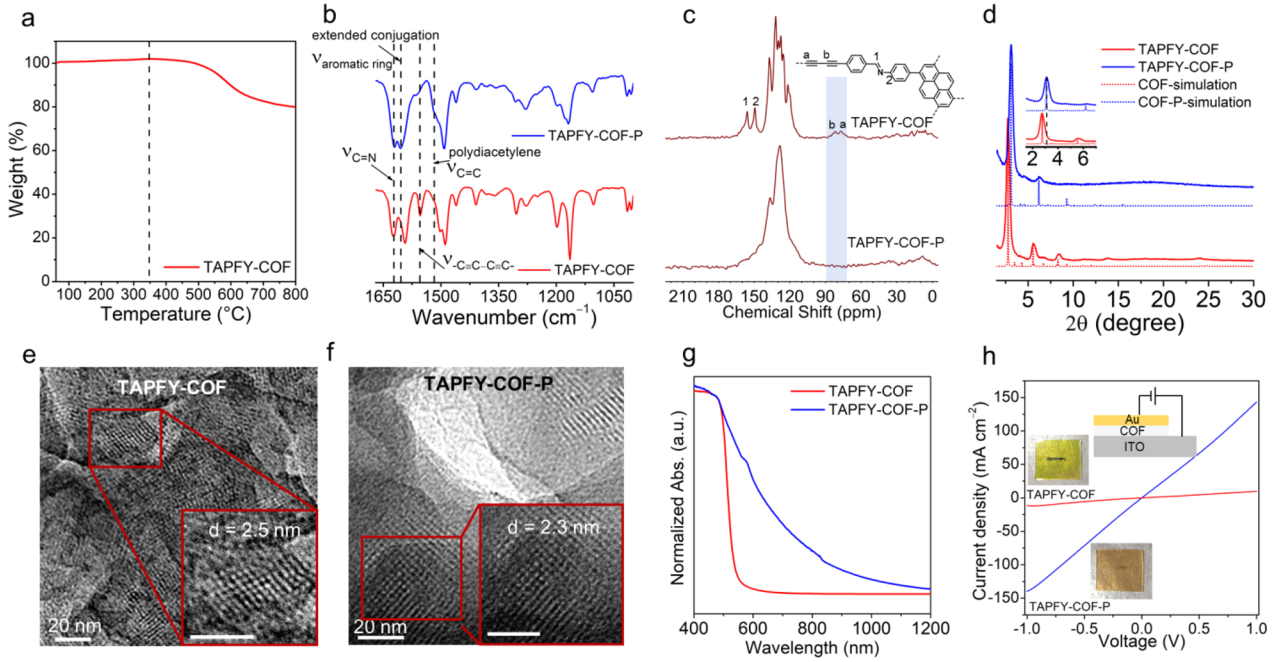


Figure 2. (a) TGA curve of TAPFY-COF. (b, c) FT-IR and ^{13}C ssNMR spectra of TAPFY-COF and TAPFY-COF-P, respectively. (d) PXRD patterns of TAPFY-COF and TAPFY-COF-P: experimentally observed (solid line) and simulated based on eclipsed stacking modes (dashed line). (e, f) TEM images of TAPFY-COF and TAPFY-COF-P, respectively. (g) UV−vis spectra of TAPFY-COF and TAPFY-COF-P. (h) *I*−*V* curves of TAPFY-COF film and TAPFY-COF-P film (inset: photographs of the films and schematic representation of conductivity measurement).

COF-P retains high crystallinity and its PXRD patterns below 10° shift to a higher angle, which is consistent with the simulated pattern of the optimized crystal structure (Figure 2d, Figure S12). We also constructed the other possible crystal model (TAPFY-COF-P_{cc}) based on different connection types, yet their PXRD simulations do not match the experimental results (Figure S13). The existence of a periodic porous framework in the two COFs and the shrinkage of the unit cell after SSTP can be observed in high-resolution transmission electron microscopy (HRTEM) (Figure 2e,f, Figure S14). N_2 sorption measurements show typical type IV isotherms for both TAPFY-COF and TAPFY-COF-P, which means they both possess mesoporous structures. Brunauer−Emmett−Teller (BET) calculation gives the surface areas of 2054 and 1030 $m^2\ g^{-1}$ for TAPFY-COF and TAPFY-COF-P, respectively, indicating that open channels are maintained. Pore size distributions calculated by non-local density functional theory (NLDFT) show that the pore size of TAPFY-COF-P is decreased (Figure S15), which is in agreement with the tendency indicated by the PXRD and HRTEM results.

To further prove the success of the topological polymerization, we prepared a diacetylene-based amorphous porous conjugated microporous polymer (TAPM-Amorphous), a small molecular model compound (DEBD-Model), and a mono-yne-based 2D COF (Py-2PE-COF) (Supporting Information). The PXRD, FT-IR, and UV−vis spectra do not show obvious changes after heating (350 °C, 6 h) for TAPM-Amorphous and Py-2PE-COF. Such changes have been found in the FT-IR and ^{13}C CP/MAS ssNMR spectra of the DEBD-Model after diacetylene polymerization.

These results prove that the stacked layer structure with proper diacetylene arrangement is necessary to conduct SSTP and that other groups or carbonization are not involved in the heating process (Figures S18−S20).

As a proof-of-concept, we prepared TAPFY-COF and TAPFY-COF-P films on conductive ITO substrate to minimize the grain boundary resistance between the particles. The conductivities of COF films in the vertical direction were measured via a two-probe strategy. Linear current−voltage profiles are obtained, indicating an ohmic contact (Figure 2h). The conductivity of TAPFY-COF-P achieves 2.8×10^{-4} S/cm, which is one magnitude higher than that of TAPFY-COF (2.2×10^{-5} S/cm) and rivals the values obtained for the best undoped COF-based materials. The enhancement in the conductivity realized by covalent conjugation of the polymeric layers can avoid the problems brought by doping with iodine or other redox-active molecules, such as the occupation of the pores, irreversible structure damage, or occasionally undesired de-doping.

In order to illustrate the generality and introduce more functional groups, we synthesized TAPB-COF and TAPP-COF, respectively. SSTP processes were also conducted at 350 °C for 6 h. FT-IR, ^{13}C CP/MAS ssNMR, EA, PXRD, and nitrogen sorption all proved the successful preparation of the two COFs and subsequent thermally induced interlayer polymerization (Figure 3 and 4a, Figures S21−S33). After polymerization, a broad absorption in the range of 700−1200 nm emerges for TAPP-COF-P, which endows it with photothermal conversion as well as photoacoustic ability triggered by near-infrared light (Figure 4b,c, Figures S36 and S37).

In addition, the unique structural features of COFs impart them with promising potential for electrocatalysis, yet they suffer from low conductivity. To reveal the advantages of our strategy, we metallized the porphyrin-based COF powder and film with Co(II) to perform a preliminary oxygen reduction reaction (Figure S39). Compared with CoTAPP-COF film, the current density of CoTAPP-COF-P film is remarkably

https://doi.org/10.1021/jacs.1c02932
J. Am. Chem. Soc. 2021, 143, 7897−7902

Downloaded from pubs.acs.org/jacsat/article-pdf/143/21/7897/3884079/ja1c02932.pdf by UNIV DE VALENCIA user on 15 October 2026

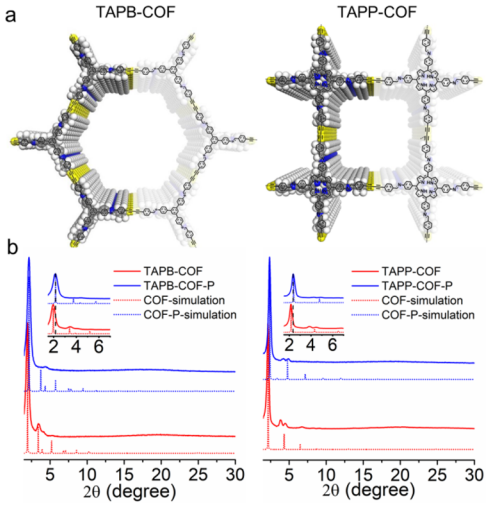


Figure 3. (a) Schematic representation of TAPB-COF and TAPP-COF. (b) PXRD patterns of TAPB-COF, TAPP-COF, TAPB-COF-P, and TAPP-COF-P: experimentally observed (solid line) and simulated based on eclipsed stacking modes (dashed line).

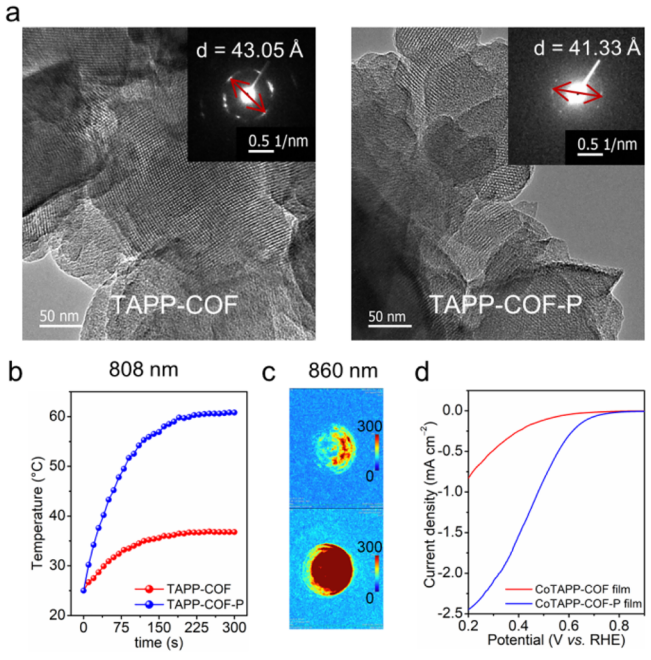


Figure 4. (a) TEM images of TAPP-COF and TAPP-COF-P. (b) Temperature change vs time for the dispersions of PEG-coated COFs. (c) Photoacoustic images of PEG-coated TAPP-COF (top) and TAPP-COF-P (bottom). (d) LSV plots of CoTAPP-COF film and CoTAPP-COF-P film.

increased, and the overpotential is decreased by 125 mV without destroying their fine chemical structure (Figure 4d).

In summary, benefitting from the unique stacking of 2D COFs, we have successfully constructed interlayer conjugated links via an SSTP in three diacetylene group-contained COFs with different topologies. Their optical and electronic properties, such as conductivity, NIR sorption, photothermal conversion, photoacoustic ability, and ORR performance, have been significantly improved after SSTP without losing their crystallinity and open channels. We expect this strategy, combined with the rich toolbox of reticular chemistry, to promote the development of COFs in optoelectronic devices as well as photo- or electrocatalytic applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.1c02932.

Materials and methods, characterization data, and additional Figures S1−S40 and Tables S1−S9 (PDF)
Crystal structure data of TAPFY-COF (CIF)
Crystal structure data of TAPFY-COF-P (CIF)
Crystal structure data of TAPB-COF (CIF)
Crystal structure data of TAPB-COF-P (CIF)
Crystal structure data of TAPP-COF (CIF)
Crystal structure data of TAPP-COF-P (CIF)

■ AUTHOR INFORMATION

Corresponding Author

Xiao Feng − *Frontiers Science Center for High Energy Material, Advanced Technology Research Institute (Jinan), Beijing Key Laboratory of Photoelectronic/Electrophotonic Conversion Materials, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, P. R. China;* orcid.org/0000-0002-3212-3051; Email: fengxiao86@bit.edu.cn

Authors

Yuhao Zhu − *Frontiers Science Center for High Energy Material, Advanced Technology Research Institute (Jinan), Beijing Key Laboratory of Photoelectronic/Electrophotonic Conversion Materials, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, P. R. China*

Pengpeng Shao − *Frontiers Science Center for High Energy Material, Advanced Technology Research Institute (Jinan), Beijing Key Laboratory of Photoelectronic/Electrophotonic Conversion Materials, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, P. R. China*

Linyu Hu − *Frontiers Science Center for High Energy Material, Advanced Technology Research Institute (Jinan), Beijing Key Laboratory of Photoelectronic/Electrophotonic Conversion Materials, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, P. R. China*

Chao Sun − *Frontiers Science Center for High Energy Material, Advanced Technology Research Institute (Jinan), Beijing Key Laboratory of Photoelectronic/Electrophotonic Conversion Materials, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, P. R. China*

Jie Li − *Frontiers Science Center for High Energy Material, Advanced Technology Research Institute (Jinan), Beijing Key Laboratory of Photoelectronic/Electrophotonic Conversion Materials, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, P. R. China*

Bo Wang − *Frontiers Science Center for High Energy Material, Advanced Technology Research Institute (Jinan), Beijing Key Laboratory of Photoelectronic/Electrophotonic Conversion Materials, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, P. R. China;* orcid.org/0000-0001-9092-3252

Complete contact information is available at:

Downloaded from pubs.acs.org/jacsat/article-pdf/143/21/7897/3884079/ja1c02932.pdf by UNIV DE VALENCIA user on 15 October 2026

https://doi.org/10.1021/jacs.1c02932
J. Am. Chem. Soc. 2021, 143, 7897−7902

https://pubs.acs.org/10.1021/jacs.1c02932

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was financially supported by the National Natural Science Foundation of China (Grant Nos. 21922502, 21674012, 21625102, 21971017, and 21490570), National Key Research and Development Program of China (2020YFB-1506300), Beijing Municipal Science and Technology Project (Z201100007520005), Beijing Institute of Technology Research Fund Program and Analysis, and Testing Center of Beijing Institute of Technolgy. We thank Dr. Donglei Chen and Prof. Chaojian Hou (Beijing Institute of Technology) for help with device fabrication and Prof. Jia Guo (Fudan University) for help with the photothermal conversion measurement.

■ REFERENCES

(1) Côté, A. P.; Benin, A. I.; Ockwig, N. W.; Keeffe, M.; Matzger, A. J.; Yaghi, O. M. Porous, Crystalline, Covalent Organic Frameworks. *Science* **2005**, *310* (5751), 1166−1170.

(2) Feng, X.; Ding, X. S.; Jiang, D. L. Covalent Organic Frameworks. *Chem. Soc. Rev.* **2012**, *41* (18), 6010−6022.

(3) Ding, S. Y.; Wang, W. Covalent Organic Frameworks (COFs): from Design to Applications. *Chem. Soc. Rev.* **2013**, *42* (2), 548−568.

(4) Huang, N.; Wang, P.; Jiang, D. L. Covalent Organic Frameworks: A Materials Platform for Structural and Functional Designs. *Nat. Rev. Mater.* **2016**, *1*, 16086.

(5) Diercks, C. S.; Yaghi, O. M. The Atom, the Molecule, and the Covalent Organic Framework. *Science* **2017**, *355* (6328), eaal1585.

(6) Lohse, M. S.; Bein, T. Covalent Organic Frameworks: Structures, Synthesis, and Applications. *Adv. Funct. Mater.* **2018**, *28* (33), 1705553.

(7) Kandambeth, S.; Dey, K.; Banerjee, R. Covalent Organic Frameworks: Chemistry beyond the Structure. *J. Am. Chem. Soc.* **2019**, *141* (5), 1807−1822.

(8) Geng, K.; He, T.; Liu, R.; Dalapati, S.; Tan, K. T.; Li, Z.; Tao, S.; Gong, Y.; Jiang, Q.; Jiang, D. Covalent Organic Frameworks: Design, Synthesis, and Functions. *Chem. Rev.* **2020**, *120* (16), 8814−8933.

(9) Huang, X.; Sun, C.; Feng, X. Crystallinity and Stability of Covalent Organic Frameworks. *Sci. China: Chem.* **2020**, *63*, 1367−1390.

(10) Hu, J. Y.; Gupta, S. K.; Ozdemir, J.; Beyzavi, M. H. Applications of Dynamic Covalent Chemistry Concept toward Tailored Covalent Organic Framework Nanomaterials: A Review. *ACS Appl. Nano Mater.* **2020**, *3* (7), 6239−6269.

(11) Martínez-Abadía, M.; Mateo-Alonso, A. Structural Approaches to Control Interlayer Interactions in 2D Covalent Organic Frameworks. *Adv. Mater.* **2020**, *32* (40), 2002366.

(12) Pang, Z. F.; Xu, S. Q.; Zhou, T. Y.; Liang, R. R.; Zhan, T. G.; Zhao, X. Construction of Covalent Organic Frameworks Bearing Three Different Kinds of Pores through the Heterostructural Mixed Linker Strategy. *J. Am. Chem. Soc.* **2016**, *138* (14), 4710−4713.

(13) Halder, A.; Karak, S.; Addicoat, M.; Bera, S.; Chakraborty, A.; Kunjattu, S. H.; Pachfule, P.; Heine, T.; Banerjee, R. Ultrastable Imine-Based Covalent Organic Frameworks for Sulfuric Acid Recovery: An Effect of Interlayer Hydrogen Bonding. *Angew. Chem., Int. Ed.* **2018**, *57* (20), 5797−5802.

(14) Jiang, L. C.; Tian, Y. Y.; Sun, T.; Zhu, Y. L.; Ren, H.; Zou, X. Q.; Ma, Y. H.; Meihaus, K. R.; Long, J. R.; Zhu, G. S. A Crystalline Polyimide Porous Organic Framework for Selective Adsorption of Acetylene over Ethylene. *J. Am. Chem. Soc.* **2018**, *140* (46), 15724−15730.

(15) Li, X.; Xu, H. S.; Leng, K.; Chee, S. W.; Zhao, X.; Jain, N.; Xu, H.; Qiao, J.; Gao, Q.; Park, I. H.; Quek, S. Y.; Mirsaidov, U.; Loh, K. P. Partitioning the Interlayer Space of Covalent Organic Frameworks by Embedding Pseudorotaxanes in Their Backbones. *Nat. Chem.* **2020**, *12*, 1115−1122.

(16) Liang, R. R.; A, R. H.; Xu, S. Q.; Qi, Q. Y.; Zhao, X. Fabricating Organic Nanotubes through Selective Disassembly of Two-Dimensional Covalent Organic Frameworks. *J. Am. Chem. Soc.* **2020**, *142* (1), 70−74.

(17) Fu, Q.; Wang, T.; Sun, Y. N.; Zheng, N.; Xie, Z. Q.; Lu, D.; Xu, Z. Y.; Wan, X. J.; Zhang, Y. M.; Liu, Y. S. A Solution-processed Nanoscale COF-like Material towards Optoelectronic Applications. *Sci. China: Chem.* **2021**, *64*, 82−91.

(18) Ding, S. Y.; Gao, J.; Wang, Q.; Zhang, Y.; Song, W. G.; Su, C. Y.; Wang, W. Construction of Covalent Organic Framework for Catalysis: Pd/COF-LZU1 in Suzuki-Miyaura Coupling Reaction. *J. Am. Chem. Soc.* **2011**, *133* (49), 19816−19822.

(19) Keller, N.; Bein, T. Optoelectronic Processes in Covalent Organic Frameworks. *Chem. Soc. Rev.* **2021**, *50*, 1813.

(20) Kuc, A.; Springer, M. A.; Batra, K.; Juarez-Mosqueda, R.; Woll, C.; Heine, T. Proximity Effect in Crystalline Framework Materials: Stacking-Induced Functionality in MOFs and COFs. *Adv. Funct. Mater.* **2020**, *30* (41), 1908004.

(21) Li, J.; Jing, X. C.; Li, Q. Q.; Li, S. W.; Gao, X.; Feng, X.; Wang, B. Bulk COFs and COF Nanosheets for Electrochemical Energy Storage and Conversion. *Chem. Soc. Rev.* **2020**, *49* (11), 3565−3604.

(22) Li, M.; Liu, J. J.; Zhang, T.; Song, X. Y.; Chen, W. H.; Chen, L. 2D Redox-Active Covalent Organic Frameworks for Supercapacitors Design, Synthesis, and Challenges. *Small* **2021**, 2005073.

(23) Liu, R. Y.; Tan, K. T.; Gong, Y. F.; Chen, Y. Z.; Li, Z. E.; Xie, S. L.; He, T.; Lu, Z.; Yang, H.; Jiang, D. L. Covalent Organic Frameworks: An Ideal Platform for Designing Ordered Materials and Advanced Applications. *Chem. Soc. Rev.* **2021**, *50* (1), 120−242.

(24) Bhadra, M.; Kandambeth, S.; Sahoo, M. K.; Addicoat, M.; Balaraman, E.; Banerjee, R. Triazine Functionalized Porous Covalent Organic Framework for Photo-organocatalytic E−Z Isomerization of Olefins. *J. Am. Chem. Soc.* **2019**, *141* (15), 6152−6156.

(25) Bi, S.; Yang, C.; Zhang, W. B.; Xu, J. S.; Liu, L. M.; Wu, D. Q.; Wang, X. C.; Han, Y.; Liang, Q. F.; Zhang, F. Two-Dimensional Semiconducting Covalent Organic Frameworks via Condensation at Arylmethyl Carbon Atoms. *Nat. Commun.* **2019**, *10*, 2467.

(26) Jakowetz, A. C.; Hinrichsen, T. F.; Ascherl, L.; Sick, T.; Calik, M.; Auras, F.; Medina, D. D.; Friend, R. H.; Rao, A.; Bein, T. Excited-State Dynamics in Fully Conjugated 2D Covalent Organic Frameworks. *J. Am. Chem. Soc.* **2019**, *141*, 11565−11571.

(27) Vitaku, E.; Gannett, C. N.; Carpenter, K. L.; Shen, L. X.; Abruña, H. c. D.; Dichtel, W. R. Phenazine-Based Covalent Organic Framework Cathode Materials with High Energy and Power Densities. *J. Am. Chem. Soc.* **2020**, *142* (1), 16−20.

(28) Yang, X. Y.; Hu, Y. M.; Dunlap, N.; Wang, X. B.; Huang, S. F.; Su, Z. P.; Sharma, S.; Jin, Y. H.; Huang, F.; Wang, X. R.; Lee, S. H.; Zhang, W. A Truxenone-based Covalent Organic Framework as an All-Solid-State Lithium-Ion Battery Cathode with High Capacity. *Angew. Chem., Int. Ed.* **2020**, *59* (46), 20385−20389.

(29) Jin, E. Q.; Asada, M.; Xu, Q.; Dalapati, S.; Addicoat, M. A.; Brady, M. A.; Xu, H.; Nakamura, T.; Heine, T.; Chen, Q. H.; Jiang, D. L. Two-Dimensional sp^2 Carbon−Conjugated Covalent Organic Framework. *Science* **2017**, *357* (6352), 673−676.

(30) Haldar, S.; Chakraborty, D.; Roy, B.; Banappanavar, G.; Rinku, K.; Mullangi, D.; Hazra, P.; Kabra, D.; Vaidhyanathan, R. Anthracene-Resorcinol Derived Covalent Organic Framework as Flexible White Light Emitter. *J. Am. Chem. Soc.* **2018**, *140* (41), 13367−13374.

(31) Li, X.; Gao, Q.; Wang, J. F.; Chen, Y. F.; Chen, Z.-H.; Xu, H.-S.; Tang, W.; Leng, K.; Ning, G.-H.; Wu, J. S.; Xu, Q.-H.; Quek, S. Y.; Lu, Y. X.; Loh, K. P. Tuneable Near White-emissive Two-Dimensional Covalent Organic Frameworks. *Nat. Commun.* **2018**, *9*, 2335.

(32) Mi, Z.; Yang, P.; Wang, R.; Unruangsri, J.; Yang, W.; Wang, C. C.; Guo, J. Stable Radical Cation-Containing Covalent Organic Frameworks Exhibiting Remarkable Structure-Enhanced Photothermal Conversion. *J. Am. Chem. Soc.* **2019**, *141* (36), 14433−14442.

Downloaded from pubs.acs.org/jacsat/article-pdf/143/21/7897/3884079/ja1c02932.pdf by UNIV DE VALENCIA user on 15 October 2026

(33) Wang, L. L.; Zeng, C.; Xu, H.; Yin, P. C.; Chen, D. C.; Deng, J. A.; Li, M.; Zheng, N.; Gu, C.; Ma, Y. G. A Highly Soluble, Crystalline Covalent Organic Framework Compatible with Device Implementation. *Chem. Sci.* **2019**, *10* (4), 1023−1028.
(34) Lakshmi, V.; Liu, C. H.; Rajeswara Rao, M.; Chen, Y. L.; Fang, Y.; Dadvand, A.; Hamzehpoor, E.; Sakai-Otsuka, Y.; Stein, R. S.; Perepichka, D. F. A Two-Dimensional Poly(azatriangulene) Covalent Organic Framework with Semiconducting and Paramagnetic States. *J. Am. Chem. Soc.* **2020**, *142* (5), 2155−2160.
(35) Prins, P.; Grozema, F. C.; Schins, J. M.; Patil, S.; Scherf, U.; Siebbeles, L. D. A. High Intrachain Hole Mobility on Molecular Wires of Ladder-Type Poly(p-Phenylenes). *Phys. Rev. Lett.* **2006**, *96* (14), 146601.
(36) Tsao, H. N.; Cho, D. M.; Park, I.; Hansen, M. R.; Mavrinskiy, A.; Yoon, D. Y.; Graf, R.; Pisula, W.; Spiess, H. W.; Mullen, K. Ultrahigh Mobility in Polymer Field-Effect Transistors by Design. *J. Am. Chem. Soc.* **2011**, *133* (8), 2605−2612.
(37) Tseng, H.-R.; Ying, L.; Hsu, B. B. Y.; Perez, L. A.; Takacs, C. J.; Bazan, G. C.; Heeger, A. J. High Mobility Field Effect Transistors Based on Macroscopically Oriented Regioregular Copolymers. *Nano Lett.* **2012**, *12* (12), 6353−6357.
(38) Cohn, G. Reactions in the Solid State. *Chem. Rev.* **1948**, *42* (3), 527−579.
(39) Baughman, R. H. Solid-State Synthesis of Large Polymer Single Crystals. *J. Polym. Sci., Polym. Phys. Ed.* **1974**, *12* (8), 1511−1535.
(40) Wegner, G. Topochemical Polymerization of Monomers with Conjugated Triple Bonds. *Makromol. Chem.* **1972**, *154* (1), 35−48.
(41) Xu, R.; Gramlich, V.; Frauenrath, H. Alternating Diacetylene Copolymer Utilizing Perfluorophenyl-Phenyl Interactions. *J. Am. Chem. Soc.* **2006**, *128* (16), 5541−5547.
(42) Terada, H.; Imai, H.; Oaki, Y. Visualization and Quantitative Detection of Friction Force by Self-Organized Organic Layered Composites. *Adv. Mater.* **2018**, *30* (27), e1801121.
(43) Ghosh, S.; Kouame, N. A.; Ramos, L.; Remita, S.; Dazzi, A.; Deniset-Besseau, A.; Beaunier, P.; Goubard, F.; Aubert, P. H.; Remita, H. Conducting Polymer Nanostructures for Photocatalysis under Visible Light. *Nat. Mater.* **2015**, *14*, 505−511.
(44) Tian, S. D.; Li, H. Z.; Li, Z.; Tang, H. J.; Yin, M. M.; Chen, Y. G.; Wang, S.; Gao, Y. T.; Yang, X. L.; Meng, F. L.; Lauher, J. W.; Wang, P.; Luo, L. Polydiacetylene-based Ultrastrong Bioorthogonal Raman Probes for Targeted Live-Cell Raman Imaging. *Nat. Commun.* **2020**, *11* (1), 81.
(45) Huang, N.; Ding, X. S.; Kim, J.; Ihee, H.; Jiang, D. L. A Photoresponsive Smart Covalent Organic Framework. *Angew. Chem., Int. Ed.* **2015**, *54* (30), 8704−8707.
(46) Jadhav, T.; Fang, Y.; Liu, C.-H.; Dadvand, A.; Hamzehpoor, E.; Patterson, W.; Jonderian, A.; Stein, R. S.; Perepichka, D. F. Transformation between 2D and 3D Covalent Organic Frameworks via Reversible [2 + 2] Cycloaddition. *J. Am. Chem. Soc.* **2020**, *142* (19), 8862−8870.
(47) Hema, K.; Ravi, A.; Raju, C.; Pathan, J. R.; Rai, R.; Sureshan, K. M. Topochemical Polymerizations for the Solid-State Synthesis of Organic Polymers. *Chem. Soc. Rev.* **2021**, *50*, 4062.
(48) Fonari, A.; Sutton, C., Validation of the Effective Masses Calculated Using Finite Difference Method on A Five-Point Stencil for Inorganic and Organic Semiconductors. *arXiv Preprint* **2013**, arXiv:1302.4996 [cond-mat.mtrl-sci].
(49) Green, M. A. Intrinsic Concentration, Effective Densities of States, and Effective Mass in Silicon. *J. Appl. Phys.* **1990**, *67* (6), 2944−2954.
(50) Inoue, K.; Koga, N.; Iwamura, H. An Approach to Organic Ferromagnets. Synthesis and Characterization of 1-Phenyl-1,3-butadiyne Polymers Having a Persistent Nitroxide Group on the Phenyl Ring. *J. Am. Chem. Soc.* **1991**, *113* (26), 9803−9810.
(51) Hu, S. L.; Zhou, Y. F.; Xue, C. R.; Yang, J. L.; Chang, Q. A Solid Reaction Towards in situ Hybridization of Carbon Dots and Conjugated Polymers for Enhanced Light Absorption and Conversion. *Chem. Commun.* **2017**, *53* (68), 9426−9429.
(52) Nishihara, H.; Hirota, T.; Matsuura, K.; Ohwada, M.; Hoshino, N.; Akutagawa, T.; Higuchi, T.; Jinnai, H.; Koseki, Y.; Kasai, H.; Matsuo, Y.; Maruyama, J.; Hayasaka, Y.; Konaka, H.; Yamada, Y.; Yamaguchi, S.; Kamiya, K.; Kamimura, T.; Nobukuni, H.; Tani, F. Synthesis of Ordered Carbonaceous Frameworks from Organic Crystals. *Nat. Commun.* **2017**, *8*, 109.
(53) Kuriyama, K.; Kikuchi, H.; Kajiyama, T. Chromatic Phase of Polydiacetylene Langmuir-Blodgett Film. *Langmuir* **1998**, *14* (5), 1130−1138.
(54) Takeoka, Y.; Asai, K.; Rikukawa, M.; Sanui, K. Incorporation of Conjugated Polydiacetylene Systems into Organic−Inorganic Quantum-Well Structures. *Chem. Commun.* **2001**, *24*, 2592−2593.
(55) Sun, X. M.; Chen, T.; Huang, S. Q.; Li, L.; Peng, H. S. Chromatic Polydiacetylene with Novel Sensitivity. *Chem. Soc. Rev.* **2010**, *39* (11), 4244−4257.

Downloaded from pubs.acs.org/jacsat/article-pdf/143/21/7897/3884079/ja1c02932.pdf by UNIV DE VALENCIA user on 15 October 2026

https://doi.org/10.1021/jacs.1c02932
J. Am. Chem. Soc. 2021, 143, 7897−7902